

FlexiKeys Product Spec — Intake, Play Check and the Five Outputs

For: CTO and main developer **From:** Founders **Date:** 6 October 2026

FlexiKeys learns about a child from three sources — the parent, the therapist and the child's own play — and turns that into five things a family uses every day: an exercise roadmap with videos, a Home Kit of tools, a daily task, nearby help with booking, and an AI assistant available 24/7.

This spec is the single source of truth for the CTO and main developer. Every decision in it was confirmed by the founders on 6 October 2026, unless it is marked **Proposal** (our recommendation, not yet confirmed) or listed under **Still open**. Where this spec and older documents disagree, this spec wins.

Two rules sit above everything else:

- 1. **No diagnosis, no scores for parents.** FlexiKeys is an education and home-practice product. It never tells a family what condition a child has, never shows a cognitive or motor "score", and never compares children.
- 2. **Safety beats convenience.** A question is required when the roadmap cannot be made safe without it. Every exercise video is approved by a physiotherapist before any family sees it. A therapist's restriction always overrides everything else.

1 Product at a glance

	Decision
Children served	About 2 to 20 years old
Conditions designed for first	Cerebral palsy, autism, speech delay
Market	Uzbekistan only for the first 2 years
Rollout	Tashkent for the first 2–3 months, then all regions of Uzbekistan, later Central Asia
App languages	Uzbek (Latin), Uzbek (Cyrillic), Russian, English — all four at launch
Sign-in	Phone number + SMS code only
Who uses it	Parents (main user), the child (games and exercises), therapists (dashboard)
Plans	Free and Pro, with a 7-day free trial of Pro
Home Kit	One standard box for every family now; customised boxes later. Priced separately from Pro
Payments	Through Telegram for now; Payme and Click later
First thing to build	The exercise roadmap with videos
Hosting	Contabo

Note on age range. 2 to 20 is very wide. The current games, the cloud mascot and the tone are designed for young children. Teenagers and young adults will need their own look, wording and exercise levels; this spec treats that as later work.

2 Three inputs, five outputs

Everything we collect builds one **child profile**. The profile drives five outputs.

Inputs

Input	What it gives	When
Parent	Intake answers, goals, daily time, "in your own words" note, optional movement videos, daily question, exercise feedback	At sign-up, daily, and again every 3 months
Therapist	Confirmed levels (GMFCS, MACS, CFCS), restrictions, goals, assigned or excluded exercises, video review	When the parent invites them, then ongoing
Games	Access settings (how the child touches), skill levels in six areas, session history	Play check at the start, then every session

Outputs

#	Output	Built from	What the family gets
1	Exercise roadmap	Goals, movement level, safety answers, therapist restrictions, daily time, skill levels	A 4-week plan, half exercises and half games, with an AI-made video for every exercise. Renewed every 4 weeks
2	Home Kit	Now: standard. Later: hand and movement answers, goals, therapist choice	A box of tools for home practice, each used in roadmap exercises
3	Daily task	Today's roadmap step, daily time, tools in the box	Today's exercises and games, marked done with one tap
4	Find help and booking	The need the parent picks, their district, their own answers	Verified clinics and specialists with approximate prices, booked in a live calendar, in person or online
5	AI assistant 24/7	The whole profile, roadmap and history	Voice-and-text help for this specific child, any time, in the parent's language

How they connect. The Home Kit decides which exercises are possible. Finished daily tasks move the roadmap forward and update skill levels. The assistant explains the roadmap and answers "how do I do this?". When the roadmap or the assistant meets something the app cannot help with, it points the family to booking. Every 3 months the parent re-answers the intake, and the next roadmap is built from the new answers.

3 The full journey

From installing the app to the first daily task takes about 15–20 minutes: around 7 minutes of questions, an 8-minute play check, and a few screens of setup. The parent can save and continue the intake later.

Step	Screen	Who	Required?	Time
1	Choose language	Parent	Required	10 s
2	Phone number + SMS code (filled in automatically)	Parent	Required	30 s
3	Basic consent, city and district	Parent	Required	30 s
4	Child basics (name, birthday, boy/girl, relationship, who fills in)	Parent	Required	40 s

Step	Screen	Who	Required?	Time
5	Health consent	Parent	Required for the exercise roadmap	20 s
6	Diagnosis and movement	Parent	Required	40 s
7	Safety questions (seizures, restrictions, recent surgery)	Parent	Required	40 s
8	Everyday abilities: talking (5 questions)	Parent	Required, "Not sure" allowed	1 min
9	Everyday abilities: hands and moving (5)	Parent	Required, "Not sure" allowed	1 min
10	Everyday abilities: seeing, hearing, daily life (5)	Parent	Required, "Not sure" allowed	1 min
11	Goals and daily time	Parent	Required	40 s
12	Routine and interests	Parent	Optional	30 s
13	In your own words (text or voice)	Parent or therapist	Optional	1–2 min
14	Movement videos	Parent	Optional	2 min
15	Invite a therapist	Parent	Optional	20 s
16	Play check intro	Parent	—	20 s
17	Play check: hands and touch	Child	Required (can retry within 3 days)	2 min
18	Play check: 4 placement games	Child	Required (can retry within 3 days)	6 min
19	The plan is ready	Parent	—	30 s
20	Free or Pro (7-day trial)	Parent	Required choice	20 s
21	Dashboard: today's task	Parent and child	—	daily

The child's own screens (play check, games) have no back buttons or menus; a grown-up leaves them by holding the pause button.

4 Sign-up and consents

- **Language first.** Four big buttons: O'zbekcha (Lotin), Ўзбекча (Кирилл), Русский, English. Changeable later in Settings. The choice sets the app text, video voice-overs and the assistant's language.
- **Phone + SMS code only.** No passwords, no Telegram, Google or Apple login. On Android the code is read from the SMS and filled in automatically. Resend after 60 seconds.
- **City and district at sign-up.** Required. Used for Find help, booking and Home Kit delivery.
- **One parent account, one or more children.** Each child has their own profile, roadmap and play check.
PROPOSAL up to 3 children per account; a second parent can join as a helper.

Consents

Consent	When it appears	Required?	If declined
Basic (account, play data)	At sign-up	Required	Cannot use the app
Health data	Before the diagnosis and safety questions	Required for the exercise roadmap	Games and Find help only; no physical exercises, because they cannot be made safe without health answers
Movement videos	Only when the parent adds a video	Required for that feature	Videos cannot be added
Therapist share	Each time the parent shares with a therapist	Required for that share	Therapist sees nothing

Each consent is one short screen in the parent's language, written in plain words, with a link to the full text.

5 Parent intake

Rule: a question is required only if the roadmap or the child's safety cannot work without it. Every required question except name, birthday, district and language has a **"Not sure"** answer, so a parent is never stuck. Answers can be saved and finished later. Parents re-answer the intake **every 3 months**; the app shows last time's answers so only changes need tapping.

Count: 29 required answers (about 7 minutes) and 12 optional ones, plus the material-allergy question when ordering a Home Kit.

A. Child and family

ID	Question as the parent sees it	Answer options	Required?	Why we need it
P1	Child's first name	Text	Required	Personalises games, videos and the assistant
P2	Date of birth	Date	Required	Content level; age limits on some exercises
P3	Boy or girl	Boy / girl	Required	Correct grammar in Uzbek and Russian voice-overs
P4	You are the child's...	Mother / father / grandparent / other carer	Required	Tone of messages; who gets reminders
P5	Who is filling this in?	Parent / therapist / together	Required	How much weight the answers get
P6	App language	Uzbek Latin / Uzbek Cyrillic / Russian / English	Required (at sign-up)	Text, voice-overs, assistant
P7	City and district	List	Required (at sign-up)	Find help, booking, kit delivery
P8	Pick your child's cloud	5 colours	Optional	Makes the app feel like the child's own

B. Health and movement (after health consent)

ID	Question	Answer options	Required?	Why we need it
P9	What has a doctor said?	Choose any: cerebral palsy, autism, Down syndrome, speech delay, ADHD, other (text), not diagnosed yet, not sure	Required	Helps choose exercises and therapists; never shown back to the parent as a label
P10	If CP: which type?	Spastic / dyskinetic / ataxic / mixed / don't know	Optional	Exercise choice (for example, no fast stretches for dyskinetic CP)
P11	If CP: which parts of the body?	One side / mostly legs / whole body / don't know	Optional	One-sided exercises; which hand to train
P12	How does your child usually move?	Walks alone / walks but needs help on stairs or long walks / walks with a walker or hands held / sits and uses a wheelchair / needs help to sit and hold the head / not sure	Required for every child	The main safety input. For CP this is the GMFCS level I–V, so the CP level is always answered. "Not sure" = sitting exercises only until a therapist confirms
P13	Has your child ever had seizures (fits)?	Yes, in the last year / yes, earlier / never / not sure	Required	Turns off flashing effects; limits exercise intensity; assistant warning rules
P14	Has a doctor said there is anything your child must not do?	Tick any: no jumping, no lying on the stomach, no neck bending, no weight on one leg, other (text) – or "Nothing"	Required	Hard exclusions in the roadmap
P15	Surgery, cast or Botox in the last 6 months?	Yes (with date and body part) / no / not sure	Required	Pauses or limits exercises for that body part
P16	Does your child use any of these?	Walker, wheelchair, leg braces, glasses, hearing aid, communication board, none	Optional	Exercise setup; screen settings
P17	Difficulty eating or swallowing?	Yes / sometimes / no / not sure	Optional	Avoids mouth and face exercises that could be unsafe
P17a	Allergies to materials (latex, rubber, dyes)?	Yes (which) / no	Required only when ordering a Home Kit	Kit safety

C. Everyday abilities (15 questions, all required)

Answers: **Yes / Sometimes (or with help) / Not yet / Not sure**, stored as 2 / 1 / 0 / empty.

ID	Group	Question	What it changes
P18	Talking	Does your child understand simple words like "come" or "give"?	Letters and communication level
P19	Talking	Can they follow two steps: "take the ball and give it to me"?	Memory and rule-switching level
P20	Talking	Do they say words you understand?	Communication level
P21	Talking	Do they put 2–3 words together?	Communication level
P22	Talking	How do they usually tell you what they want? (words / pointing or looking / sounds / pictures)	Suggests the My Voice talking board

ID	Group	Question	What it changes
P23	Hands	Can they pick up a small toy with their fingers?	Finger exercises and kit tools
P24	Hands	Can they hold a spoon or a crayon?	Grip exercises
P25	Hands	Can they draw lines or write any letters? (letters / lines / not yet)	Tracing and letter games level
P26	Hands	Can they tap a phone screen where they want?	First guess of touch settings
P27	Hands	Can they sit on their own without support?	Exercise positions; posture advice
P28	Every day	Do they see small pictures clearly? (yes / wears glasses / not sure)	Button size and contrast
P29	Every day	Do they hear you from another room?	Adds visual cues to sounds
P30	Every day	Can they play with one toy for about 5 minutes?	Session length
P31	Every day	Do they eat by themselves?	Daily-life exercises
P32	Every day	Do they do small things alone, like taking off shoes?	Daily-life exercises

D. Goals and routine

ID	Question	Answer options	Required?	Why we need it
P33	What would you most like your child to do better in the next 3 months?	Pick 1–3: walk more steadily, use hands better, sit better, talk or communicate, understand more, eat by themselves, play with others, get ready for school	Required	The roadmap's targets
P34	How much time can you spend on exercises each day?	10 / 20 / 30 minutes	Required	How many exercises per day
P35	Best time of day for practice	Morning / afternoon / evening	Optional	Reminder time
P36	Where does your child spend the day?	Home / kindergarten / rehab centre / school	Optional	When to schedule tasks
P37	What does your child love?	Music, cars, animals, cartoons, water, balls, other	Optional	Themes for games and videos
P38	What upsets or tires your child quickly?	Text	Optional	Session length; assistant advice

E. Your own words and media

ID	Question	Answer	Required?	Why we need it
P39	Tell us about your child in your own words	Text (up to 1000 characters) or a voice note (up to 2 minutes)	Optional	Catches what fixed questions miss. AI pulls out a few hints (stronger hand, tiredness limit, interests); the parent confirms them with one tap

ID	Question	Answer	Required?	Why we need it
P40	Videos of your child moving: walking or crawling, picking up a toy, drawing or tapping	Up to 3 clips, 20–30 seconds each	Optional	Lets a therapist review movement remotely. Only a therapist the parent chooses can watch them; AI never analyses them
P41	Invite your child's therapist	Phone number or Telegram link	Optional	Unlocks the therapist intake and dashboard

6 Therapists

Who can join: physiotherapists and LFK specialists, speech therapists (logopedes), defectologists and occupational therapists, and neurologists.

How they join: the parent sends an invite by phone number or Telegram link (P41). **PROPOSAL** partner centres can also invite their own therapists. **PROPOSAL** we check each therapist's diploma by hand before their first share is active.

What they get: a therapist dashboard (with the AI assistant) showing only the children whose parents shared with them, for as long as the parent allows.

Therapist intake

ID	Question	Answer	Required in the therapist form?	Why we need it
T1	Name, profession, workplace	Physio / LFK, speech therapist, defectologist / OT, neurologist	Required	Trust, and which fields apply
T2	Gross motor level (GMFCS)	I–V	Required	Confirms or corrects P12
T3	Hand use level (MACS)	I–V	Required for physio / LFK and OT; optional for others	Hand exercise level
T4	Communication level (CFCS)	I–V	Required for speech therapists; optional for others	Speech goals and My Voice
T5	CP type and body parts	As P10–P11	Optional	Corrects parent answers
T6	What must this child not do at home?	Checklist + text, or "None"	Required	Hard exclusions; overrides P14
T7	Top goals for the next 3 months	Up to 3, from the P33 list or own words	Required	Roadmap targets; override P33
T8	Exercises to assign or exclude	From the approved video library	Optional	Direct control of the roadmap
T9	Review of the parent's movement videos	Notes per clip	Optional	Leg and balance exercises; whether leg tools go in the kit
T10	How often do you want progress updates?	Weekly / every 2 weeks / monthly	Optional	Report schedule

ID	Question	Answer	Required in the therapist form?	Why we need it
T11	Message to the parent	Text	Optional	Shown on the parent dashboard

What therapists control

- **Their answers win.** When a therapist answer differs from the parent's, the therapist's is used, and the parent sees "Updated by [therapist name]".
- **Restrictions are absolute.** Nothing in T6 can ever appear in a roadmap, a daily task or an assistant answer.
- **Leg and balance tools** go into a child's Home Kit only with therapist approval.
- **Exercise feedback.** When a parent films the child doing an exercise (optional), the therapist watches and replies in the app.
- **Assistant chats:** the therapist sees a **summary only** — the topics the parent asked about and any warnings — never the full conversation.

7 Play check

About 8 minutes of play, right after the intake. The child plays; the parent sits next to them. **There are no scores for the parent** — the result is only used to choose the starting games and their levels.

If the child is upset or tired, the parent can stop. The app offers a retry within 3 days and meanwhile starts the child on the easiest games. The parent may help during the play check; the app records that they did.

Stage 1 — Hands and touch (about 2 minutes)

No thinking content, only how the child uses the screen. This comes first because a child with CP who misses a small button may understand perfectly; if we count that as a wrong answer, we choose the wrong games for months.

Measured	How	Stored as
Input mode	Try normal touch; if 3 of 5 taps miss or repeat, offer hold-to-choose; then switch scanning	<code>input_mode</code> : touch / dwell / switch / keyboard
Smallest button that works	Targets shrink until 2 misses in a row, then step back one size	<code>min_target_mm</code> (never below 12)
Hold time	The child holds the ball as long as is comfortable	<code>dwell_ms</code> , 500–3000
Accidental repeats	A second touch within 300 ms	<code>repeat_lock</code> : on / off
Clean release	Can the child lift the finger cleanly after holding	<code>release_ok</code> : yes / no

These five values are the child's **access settings**. Every game reads them.

Stage 2 — Four placement games (about 6 minutes)

Each game at its easiest level, 3–5 tries. The only question per area: **can the child do this right now — not yet, with help, or yes?**

Area	Game	One try
Seeing and attention	Quiet Forest, 2 distractors	Find the spotted butterfly among 3
Memory	What's Missing?, 2–3 items	See a tray, one item hides, pick it from 3

Area	Game	One try
Letters	Letter Hunt, 2 options	Hear a sound, tap its letter
Thinking	Sort Two Ways, one rule	Put a card in the box with the same shape

Rules the build must keep: no timer; no red, no X, no failure sound; after 2 mistakes the right answer gently shimmers, after 4 the others fade, so every try ends in success.

How one try is recorded

```
{
  "trial_id": "uuid", "child_id": "uuid", "session_id": "uuid",
  "game": "whats_missing", "level": 1, "item_id": "tray_apple_ball_fish",
  "input_mode": "dwell", "outcome": "correct", "attempts": 2,
  "hint_shown": "none", "access_event": false, "helped_by_adult": false,
  "response_ms": 8400, "ts": "2026-10-06T10:12:03Z"
}
```

`access_event` is the most important field: a missed tap, double tap or hold released too early is marked as an access event, not a wrong answer, and is left out of the skill calculation. `response_ms` is saved but never scored.

8 Output 1 — Exercise roadmap and AI videos

This is the first thing to build and the heart of the product.

Shape of the roadmap

- **Length:** 4 weeks. At the end, a new roadmap is built from what the child completed, skill-level changes and any new answers.
- **Mix:** about half home exercises and half games, by time.
- **Daily amount** follows the parent's answer to P34: 10 minutes $\approx$ 1–2 exercises, 20 minutes $\approx$ 2–3, 30 minutes $\approx$ 4–5, with games filling the rest.
- **Child's game limit:** 20 minutes a day, then the app stops gently with the sleeping cloud.
- **Targets** come from the goals (P33, or T7 when a therapist has set them).
- **Never in a roadmap:** anything in the parent's restrictions (P14) or the therapist's (T6); exercises for a body part with recent surgery (P15); standing or walking exercises when the movement answer is "Not sure" or level IV–V; leg and balance exercises for CP unless a therapist approved them; high-intensity exercises when seizures were reported in the last year (P13).

The videos

- **Fully AI-generated**, including the person demonstrating each movement, with an **AI voice** that says the child's name in the parent's language.
- **A physiotherapist approves every video before any family sees it.** AI video can show a wrong joint angle or posture, and a wrong movement practised every day becomes a habit for a child with CP. The system needs a review queue: generated $\rightarrow$ physio review (approve / reject / request changes with a note) $\rightarrow$ published. Only approved videos can enter a roadmap. Each video stores who approved it and when.
- **Each video is tagged** with: body area, goal, movement levels it suits (GMFCS I–V), age band, Home Kit tool used, difficulty level (1–3), restrictions it conflicts with, and language versions.

- **Offline:** the next 7 days of videos download automatically on Wi-Fi so families with weak internet can still practise.
- **Length:** **PROPOSAL** 1–3 minutes per exercise video.

Parent feedback

- After each exercise the parent taps **Done**, and **PROPOSAL** optionally how it went (easy / OK / hard).
- **Optional video feedback:** the parent can film the child doing the exercise; the child's therapist watches and replies. AI does not analyse these videos.
- Two "hard" ratings in a row lower that exercise's level; two "easy" ratings raise it, unless a therapist has fixed the level.

9 Output 2 — Home Kit

Now: one standard box

Every family gets the same box of tools for home practice. Each tool has a card with a QR code that opens its how-to videos. Contents are chosen from the planned catalogue (therapy putty, tongs and pom-poms, lacing cards, big-knob pegboard, soft foam ball, bean bags and target, tactile foot discs, sequencing and emotion cards, and similar). **Still open:** the exact list for the standard box.

Later: a customised box

- A catalogue of about **30 tools**.
- Each child gets **8–12 tools** chosen for them.
- Chosen by **rules from the intake plus the therapist:** the app proposes a box from the hand and movement answers, goals and kit allergies; the therapist can change it.
- **Leg and balance tools only if a therapist approves.**
- Families can **swap or add tools** later, for example when the roadmap moves up a level.

Price and delivery

- The Home Kit has **its own price**, separate from the Pro subscription. Discounts or bundles may come later.
- Whether buying a kit includes free Pro months is **still open**.
- **Delivery to all of Uzbekistan**, by courier or pickup at a partner centre.
- **Safety:** suppliers must provide toy-safety reports (EN 71 or ASTM F963); every small part passes a choke test. Material allergies (P17a) are checked before shipping.
- **Use tracking:** scanning a tool's QR card and tapping Done counts towards the roadmap.

10 Output 3 — Daily task

The dashboard opens on **Today:** the day's exercises (each with its video and the tool it needs) and the day's games, sized to the parent's daily time.

- One tap marks each exercise done; the week view shows done, today and coming days.
- A reminder arrives at the parent's preferred time (P35). **PROPOSAL** push notification, with a Telegram option.
- If the child is sick or very tired, the parent can **pause today**; the roadmap shifts without counting it as missed. **PROPOSAL** a lighter version is offered first.
- One **daily question** (for example "Did your child point to something they wanted today?") adds small, steady updates to the profile between the 3-month re-intakes.

- Every week the parent gets two plain sentences about the week ("Aziz did 5 of 7 tasks and now holds the crayon for 10 seconds"), written by AI from the numbers only.

11 Output 4 — Find help and booking

Find help

- **About 40 verified listings at launch, Tashkent first**, then the regions as the rollout grows. Each one is checked by phone by our team.
- The parent picks a need in plain words: speech and talking / movement and hands / learning and attention / a medical check-up / I'm not sure. Their own intake answers can put the most relevant need first.
- **"I'm not sure"** never guesses: it suggests asking the child's doctor or a free state centre first.
- Results: **free state centres first**, then private places by distance.
- Each listing shows: name, type, district, map link, contacts, languages, ages served, whether they work with CP, **approximate price range and the date it was checked**. Prices older than 6 months show "price not recently checked".
- Parents can report a wrong price or a closed place; the listing is flagged within 24 hours.
- **Parent reviews and ratings: later**, once there are enough real bookings.
- **The boundary:** game scores are never used to tell a family their child needs a particular doctor. That would be a medical referral.

Booking

- **Real-time calendar from version 1.** Parents see a clinic's or specialist's free slots and book instantly.
- **In-person and online sessions** can both be booked.
- **The family pays the clinic directly.** FlexiKeys handles no session money.
- **No money from clinics for now.** Commission or clinic subscriptions will be decided after the pilot.
- Clinics need a **clinic dashboard** to publish slots, confirm or cancel bookings, and add online-session links.
- **PROPOSAL** a reminder 1 day and 2 hours before the session, and a short "how did it go?" afterwards.
- **PROPOSAL** cancellation rules are each clinic's own, shown before booking.
- With the parent's permission, the booking can include a therapist share of the child's weekly summary.

12 Output 5 — AI assistant 24/7

- **Where:** in the parent app and in the therapist dashboard.
- **How: voice in and voice out**, plus text. Parents can speak instead of typing, and hear the answer read aloud.
- **Languages:** the same four as the app.
- **What it knows:** the child's profile, current roadmap, completed tasks, the therapist's restrictions and notes, and our approved content library.
- **What it helps with:** how to do an exercise, what is next in the roadmap, what to do if the child refuses or is tired, how to use a Home Kit tool, app help, and finding or booking a specialist.
- **Medical questions:** it gives general information in plain words, then advises seeing a doctor and offers booking. It never diagnoses, never comments on medication doses, and never contradicts the therapist's restrictions.
- **Emergencies:** if the parent mentions a seizure, trouble breathing, a fall with a head injury or sudden strong pain, the assistant immediately shows **103** and the advice to call a doctor, before anything else.
- **Free plan:** about 20 messages a day. **Pro:** unlimited.

- **The therapist** sees a summary only: topics asked and any warnings, not the full chat.
- **It may suggest roadmap changes**, but the parent confirms them, and therapist restrictions still apply.
- **PROPOSAL** a therapist reviews a sample of answers every month for quality; chats are kept until the parent deletes them.

13 How the plan is calculated

The roadmap and the game choice come from a **fixed skill model and simple rules**, not from an AI guessing. That lets us explain every choice to a therapist, test it, and get the same result from the same inputs. AI only writes the friendly sentences the parent reads, and only from facts it is given.

Skill levels

Six skill areas: **access**, **attention**, **memory**, **flexibility**, **literacy**, **numeracy**. Each has one number, θ (theta).

1. **First guess from the intake.** Related answers (0–2 each) give a starting θ . A therapist's answers weigh more than a parent's.
2. **Corrected by the play check.** Each area is put in a bucket — not yet / with help / yes — using only clean tries. If the play check disagrees with the intake, the play check wins by one step at most.
3. **Updated after every clean try** with the Elo rule:

$$P(\text{correct}) = 1 / (1 + e^{-(\theta - d)})$$

$$\theta_{\text{new}} = \theta + K \times (\text{correct} - P(\text{correct})), \quad K = 1 / (1 + 0.05 \times n)$$

d is the difficulty of the game level; n is the number of tries so far. K starts large and shrinks as evidence builds, so after about a week of play the play-check guess no longer matters.

Choosing games

Slot	Chosen from	Aim to succeed
Easy win	The child's strongest area	About 85% of the time
Growth	The weakest area that has a game	About 75% of the time
Child's choice	Any unlocked game	—

- At most **one new game a week**; an unlocked game is never taken away.
- Game time counts towards the 20-minute daily limit.

Choosing exercises

1. Start from the full approved video library.
2. Remove every video that conflicts with P13–P15, P14/T6 restrictions, the movement level, age band, or a tool the family does not have.
3. Keep videos that serve the goals (T7 or P33).
4. Pick the starting difficulty from the hand and movement answers, then adjust from Done / easy / hard feedback.

- Spread them across 4 weeks to fit the daily time (P34), mixing body areas so the same muscles are not worked two days running.

Explaining the plan

The rules output structured reasons (for example `slot=growth, area=memory, game=whats_missing, level=2`). AI turns these into two or three warm sentences in the parent's language. It may only restate facts it was given; it cannot invent progress or add advice.

14 Plans, prices and payments

	Free	Pro
Price	0 so'm	Still open (per month)
Trial	—	7 days free
Intake and play check	Yes	Yes
Exercise roadmap with videos	PROPOSAL first week only	Full 4-week roadmaps
Games	PROPOSAL 3 at a time	All games as the child is ready
Offline video downloads	No	Yes
Find help and booking	Yes	Yes
AI assistant	About 20 messages a day	Unlimited
Therapist sharing and feedback videos	PROPOSAL no	Yes
Weekly summary	Yes	Yes

- Home Kit** is sold separately at its own price (**still open**). Bundles or discounts may come later.
- Payments for now go through Telegram**: the parent pays through our Telegram flow and the account is upgraded. **Payme and Click** are added later inside the app.
- PROPOSAL** prices shown in so'm only; the trial ends with a reminder 2 days before.

15 Data: uses, limits, storage and consent

What the data is used for

Use	Data it needs	Who benefits
Fit the screen to the child's hands	Access settings	Child
Build the exercise roadmap	Intake, safety answers, goals, therapist restrictions, feedback	Child, parent
Choose games and levels	Skill levels, access settings	Child
Choose Home Kit contents (later)	Hand and movement answers, goals, therapist choice, allergies	Parent, child
Suggest places and specialists	The need the parent picks, district, their own answers	Parent
Book sessions	Chosen slot, contact, optional shared summary	Parent, clinic
Answer in the AI assistant	Profile, roadmap, history	Parent, therapist

Use	Data it needs	Who benefits
Weekly summary	Session counts, skill changes	Parent
Therapist view	Summary data, through a share the parent creates	Therapist
Improve the product	Anonymous totals only	Everyone

Never allowed

- No diagnosis or triage, and no telling a family their child needs a particular doctor because of game data.
- No scores, percentages or comparisons shown to parents.
- No selling or sharing data with anyone, including clinics. A clinic sees a child only if the parent shares.
- No advertising based on a child's condition.
- No AI analysis of children's movement videos or feedback videos.
- No research use without separate, clear consent, even anonymised.

Storage

- **Hosting: Contabo.** Contabo has no data centre in Uzbekistan, so a lawyer must confirm that Uzbek personal-data law allows children's health data to be stored abroad **before real centre data goes on the servers.**
- Health answers are stored in **encrypted** fields.
- Detailed try-by-try records stay **on the phone** (last 90 days); the server keeps only skill levels and session counts.
- **Movement and feedback videos are kept until the parent deletes them.** The parent can delete any clip at any time, and stopping a therapist share removes the therapist's access at once.
- AI-generated exercise videos are our content, not personal data.

Record	Key fields	Where	Kept
parent	id, phone, language, district, relationship	Server	Until the account is deleted
child	id, name, birth_date, sex, avatar_color, respondent	Phone + server	Until deleted
health_profile	diagnoses[], cp_type, cp_parts, movement_level, seizures, restrictions[], surgery, equipment[], swallowing, allergies	Phone + server (encrypted)	Until deleted
intake_answer	question_id, value, answered_at, respondent	Phone + server	Until deleted; history of every 3-month round
goals	goals[], daily_minutes, best_time, interests[]	Phone + server	Until deleted
parent_note	text or audio, confirmed hints	Phone; hints on server	Until deleted
media	movement and feedback clips, tag, share_id	Encrypted storage	Until the parent deletes
therapist_input	T1-T11, therapist_id, updated_at	Server	While the share is active
access_profile	input_mode, min_target_mm, dwell_ms, repeat_lock, release_ok	Phone + server	Updated continuously
trial	the try record	Phone only	90 days

Record	Key fields	Where	Kept
skill_estimate	area, theta, n_trials	Phone + server	Current + weekly snapshots
roadmap	weeks[], exercises[], games[], reasons[], status	Phone + server	Current + history
video	id, tags, approved_by, approved_at, status	Server	Library content
kit_order	kit_type, items[], address, delivery, status	Server	As required for accounting
booking	clinic_id, specialist_id, slot, mode, status	Server	Until deleted
assistant_chat	messages, summary for therapist	Server	Until the parent deletes
share	child_id, therapist_id, scope, expires_at	Server	Until stopped

Parent controls

- See everything stored about the child, in plain words.
- Download it as a file.
- Delete any video, chat or the whole child profile (server copy removed within 30 days, phone copy at once).
- Stop any therapist share in one tap.

16 Build order and acceptance criteria

The exercise roadmap comes first, so the order starts with what it needs. Size: S = under 1 week, M = 1–3 weeks, L = 3–6 weeks (our estimate; engineering to correct).

#	Build	Done means	Size
1	Sign-up	Four languages; phone + SMS with auto-filled code on Android; district list; basic consent	M
2	Parent intake	All P1–P41 questions in four languages; required and optional exactly as specified; "Not sure" stored as empty; save and continue later; health-consent gate (declined = games and Find help only)	M
3	Video library + physio review queue	All videos generated per exercise and language; physio can approve, reject or request changes with a note; only approved videos are usable; every video tagged as specified	L
4	Roadmap engine	4-week roadmap, half exercises and half games; exercises per day follow P34; nothing that conflicts with P13–P15 or restrictions can ever be selected (covered by tests); same inputs give the same roadmap	L
5	Daily task + offline	Today screen; one-tap Done with optional easy / OK / hard; pause today; next 7 days of videos downloaded on Wi-Fi	M
6	Input layer + try records	Games receive select / move / hold actions, never raw taps; hold-to-choose and switch scanning work; every try saved with access events	L
7	Play check	Hands-and-touch stage finds all five access settings; 4 placement games; under 10 minutes; stop and retry within 3 days	M
8	Skill model + game choice	θ per area from intake + play check + Elo; 3-slot sessions; 20-minute daily limit; one new game a week	M

#	Build	Done means	Size
9	Plans + Telegram payments	Free and Pro with 7-day trial; upgrade after Telegram payment; Free limits enforced	M
10	Therapist dashboard	Invite, intake T1–T11, therapist answers override parent's, assign or exclude exercises, review movement and feedback videos, see assistant summaries	L
11	AI assistant	Voice in and out plus text; four languages; answers only from profile and approved content; emergency words show 103 first; 20 messages a day on Free; summary to therapist	L
12	Find help	About 40 verified Tashkent listings; free first, then distance; price check dates; report wrong info	M
13	Booking + clinic dashboard	Real-time slots; in-person and online; instant booking; clinic confirms or cancels; reminders	L
14	Home Kit orders	Standard box order with its own price; delivery across Uzbekistan by courier or pickup; QR cards open videos	M
15	Data controls	See, download and delete; encrypted health fields; video deletion; share revocation	M
16	Customised Home Kit (later)	\~30-tool catalogue; rules propose 8–12 tools; therapist edits; leg tools need approval; swaps and add-ons	L

17 Pilot measures

- Parents who finish sign-up, intake and play check: **above 80%**.
- Time to finish: **under 20 minutes**.
- Daily tasks done: **at least 4 of 7 days** for most active families.
- Play-check tries marked as access events: **above 30%** for a child means the touch settings are wrong.
- Physio review: time from AI generation to approval, and share of videos rejected.
- Therapists who agree with the first roadmap when they review it.
- Bookings made in Tashkent in the first 2–3 months.

18 Still open

- ☐ Pro price per month.
- ☐ Home Kit price, and whether a kit includes any free Pro months.
- ☐ Exact items in the standard Home Kit.
- ☐ Lawyer's answer on storing children's health data on Contabo outside Uzbekistan.
- ☐ Who the reviewing physiotherapist is, and how many videos they can approve per week.
- ☐ Pilot start date and partner centres.
- ☐ Exactly what Free includes (the table above is a proposal).
- ☐ Content and design for teenagers and young adults (ages 13–20).
- ☐ Commission or subscription for clinics, after the pilot.
- ☐ Which Telegram payment flow we use, and when Payme and Click are added.

19 Words used in this spec

Term	Meaning
Access settings	How this child best uses the screen: touch or hold, button size, hold time, protection from accidental taps
Access event	A try that went wrong because of the hand, not the mind. Not counted as a wrong answer
Hold-to-choose (dwell)	The child keeps a finger on a button for a set time to select it
Switch scanning	Options light up one by one; the child presses one big button to pick the lit one
GMFCS I–V	The 5-level scale doctors use for how a child with CP moves. I = walks alone, V = needs help to sit
MACS, CFCS	Similar 5-level scales for hand use and for communication
θ (theta)	One number per skill area showing the child's current level
Elo	The rating method from chess, used to update θ after every try
Roadmap	The child's 4-week plan of exercises and games
Restriction	Something a doctor or therapist says the child must not do. Never appears in a roadmap

20 Sources

- Founders' decisions, 6 October 2026 (60 questions answered by Charos)
- Internal: Onboarding assessment and adaptive game assignment spec (September 2026)
- Internal: Game catalogue, 3 → 20 games (September 2026)
- Internal: Home Kit tools summary (September 2026)
- Internal: FlexiKeys App Design canvas
- Pelánek, *Applications of the Elo rating system in adaptive educational systems*, Computers & Education, 2016
- Palisano et al., Gross Motor Function Classification System (GMFCS)